Theoretical Bounds on χ_ν in Observer-Patch Holography

Canonical Coherent-Matter Susceptibility and Repair-Charge Coupling

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Abstract

The coherent-material branch of Observer-Patch Holography (OPH) supplies a canonical scalar source in the same quotient-edge register as the protected collar reserve. On the co-registered presence branch, the dimensionless coupling is $\chi_\nu^{\mathrm{can}} = 1 - P/24 = 0.9320429912748350\ldots$. Under the weaker mean-count reading it lies between that value and one. This is a channel coefficient, not a force coefficient. A proposed repair-charge condensate action couples the source to a dynamical repair phase. Conditional on that action, repair-current continuity, dilute dust behavior, spherical deep-galaxy scaling, field stress, and a lawful coherent-source force follow from one variational system. A compact neutral coherence contrast has no monopole and cannot support a closed device. The dimensional source calibration, ambient repair field, compact-phase completion, and laboratory source receipt are work in progress; no numerical lift prediction follows from χ_ν alone.

Claim Boundary

This paper proves a finite-channel coefficient on a declared OPH branch. It also derives continuity, dilute dust behavior, spherical deep-galaxy scaling, force, and momentum formulae conditional on the repair-charge condensate action in the dark-sector paper [4]. The OPH derivation of that action and its dimensional source normalization are work in progress.

The exact scope is:

Object	Scope
Coherent source in the quotient-edge scalar register	finite theorem under the stated source receipts
$\chi_\nu^{\mathrm{can}} = \lambda_{\mathrm{collar}}$	theorem under co-registration and reserve-survival premises
$\chi_\nu^{\mathrm{can}} = 1 - P/24$	exact on the presence branch
Repair-charge action	proposed dynamical completion
Continuity, dust limit, and spherical deep-galaxy scaling	derived conditional on that action and its constitutive premises
Dimensional coupling $q_\star$	work in progress
Device force	conditional on source charge, external field, and momentum closure
Numerical lift	absent

Contents

1 Canonical Coherent-Matter Source

An OPH technology instantiation is an observer-like self-reading system: a bounded physical or software patch with local state, ports or boundaries, readback, records, feedback or repair moves, and a public evidence bundle. For a nondegenerate coherent material subfederation U, define the canonical source

$$S_{\mathrm{coh}}^{\mathrm{can}} = \mathbf{1}_{\text{self-read}}\, \mathsf{R}_U \mathsf{P}_U \mathsf{C}_U. \tag{1.1}$$

The factors state that the patch reads itself, produces a stable record, supports a predictive boundary, and executes a registered repair or control move. If any factor vanishes, the coherent scalar source is zero.

The screen-microphysics branch supplies four structural facts [3]:

S1. $S_{\mathrm{coh}}^{\mathrm{can}}$ descends to a protected quotient observable.

S2. A nondegenerate source is nonzero.

S3. Scalar edge-center exhaustion places it in the unique quotient-local scalar register.

S4. The protected $\mathbb{Z}_6$ reserve acts on that same register and commutes with scalar activation.

These statements identify the channel. They supply no spacetime action or dimensional gravitational charge.

2 Protected-Reserve Coefficient

Let $\epsilon_{\mathbb{Z}_6}(y)$ be the conditional reserve-presence profile on the co-registered scalar slot, and let $w(y)$ be the normalized scalar weight across the finite collar. The scalar opportunity survives with local factor

$$\lambda_{\mathrm{slot}}(y) = 1 - \epsilon_{\mathbb{Z}_6}(y). \tag{2.1}$$

The finite-thickness collar coefficient is

$$\lambda_{\mathrm{collar}} = \int \mathrm{d}y\, w(y) \left[1 - \epsilon_{\mathbb{Z}_6}(y)\right]. \tag{2.2}$$

The protected-reserve receipt gives

$$\int \mathrm{d}y\, w(y) \epsilon_{\mathbb{Z}_6}(y) = \frac{P}{24}. \tag{2.3}$$

Theorem 2.1 (Canonical susceptibility)**.** *Assume the nonzero coherent-source branch, scalar edge-center exhaustion, co-registration with the protected reserve, and the presence reading in Eqs.* (2.2)–(2.3)*. Then*

$$\boxed{\chi_\nu^{\mathrm{can}} = \lambda_{\mathrm{collar}} = 1 - \frac{P}{24}}. \tag{2.4}$$

For $P = 1.630968209403959$*,*

$$\boxed{\chi_\nu^{\mathrm{can}} = 0.9320429912748350\ldots}. \tag{2.5}$$

Proof. The coherent source enters the scalar opportunity count before the protected reserve acts. Co-registration makes the local surviving fraction $1 - \epsilon_{\mathbb{Z}_6}(y)$. Averaging with $w(y)$ gives $\lambda_{\mathrm{collar}} S_{\mathrm{coh}}^{\mathrm{can}}$. The definition of canonical susceptibility gives $\chi_\nu^{\mathrm{can}} S_{\mathrm{coh}}^{\mathrm{can}}$. Nondegeneracy permits cancellation of $S_{\mathrm{coh}}^{\mathrm{can}}$, and Eq. (2.3) gives the value. □

Corollary 2.2 (Mean-count band)**.** *If* $\epsilon_{\mathbb{Z}_6}(y)$ *is interpreted only as the mean of a nonnegative integer occupancy, then*

$$\boxed{0.9320429912748350\ldots \le \chi_\nu^{\mathrm{can}} \le 1}. \tag{2.6}$$

Proof. Markov's inequality gives $1 - \epsilon \le \Pr[N = 0] \le 1$. Integration and Eq. (2.3) give the band. □

The theorem excludes a zero scalar-channel coefficient on the stated nondegenerate branch. It does not exclude a zero dimensional gravitational coupling $q_\star$, a zero device repair charge, or a zero ambient repair gradient.

3 Engineering Coordinates

An engineering description may absorb an effective coherent capacity into the source coordinate:

$$S_{\mathrm{coh}}^{\mathrm{eng}} = N_{\mathrm{coh}} S_{\mathrm{coh}}^{\mathrm{can}}. \tag{3.1}$$

Invariance of the scalar perturbation requires

$$\chi_\nu^{\mathrm{can}} S_{\mathrm{coh}}^{\mathrm{can}} = \chi_\nu^{\mathrm{eng}} S_{\mathrm{coh}}^{\mathrm{eng}}, \qquad \boxed{\chi_\nu^{\mathrm{eng}} = \frac{\chi_\nu^{\mathrm{can}}}{N_{\mathrm{coh}}}}. \tag{3.2}$$

Very small values of χ_ν^{eng} can therefore represent an order-one canonical coefficient. Equation (3.2) is a chart identity. It does not define $q_\star$, measure $S_{\mathrm{coh}}^{\mathrm{can}}$, or predict force.

4 Repair-Charge Coupling

The dark-sector completion takes integer scalar repair occupation and a compact repair phase to a canonical pair (n, θ). On a fixed real lift of the phase, its relevant action is

$$\begin{aligned} I_{\mathrm{RPC}} = I_{\mathrm{Regge}} &+ I_{\mathrm{SM}} \\ &+ \int \mathrm{d}\tau \Bigg[-\sum_v V_v n_v D_\tau \theta_v - \sum_v V_v \varepsilon(n_v) - \sum_e W_e \mathcal{K}(|(d\theta)_e|) \\ &\qquad - \sum_v V_v \theta_v \left(\mathcal{Q}_{b,v} + q_\star \chi_\nu^{\mathrm{can}} S_{\mathrm{coh},v}^{\mathrm{can}} \right) \Bigg]. \end{aligned} \tag{4.1}$$

The source and canonical signs are chosen so phase variation gives the positive-source balance in Eq. (4.3), and translation of a positive localized source gives force $-Q_R \nabla\theta$. A globally single-valued compact-phase coupling, the derivation of this action from finite repair dynamics, and the dimensional constants are work in progress.

The coherent repair-charge density is

$$\mathcal{Q}_{\mathrm{coh}} = q_\star \chi_\nu^{\mathrm{can}} S_{\mathrm{coh}}^{\mathrm{can}}. \tag{4.2}$$

The phase equation is

$$D_\tau n + d^\dagger j_R = \mathcal{Q}_b + q_\star \chi_\nu^{\mathrm{can}} S_{\mathrm{coh}}^{\mathrm{can}}. \tag{4.3}$$

For a finite region Ω,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \sum_{v\in\Omega} V_v n_v + \sum_{e\in\partial\Omega} W_e j_{R,e} = \sum_{v\in\Omega} V_v \left(\mathcal{Q}_{b,v} + q_\star \chi_\nu^{\mathrm{can}} S_{\mathrm{coh},v}^{\mathrm{can}} \right). \tag{4.4}$$

The boundary flux is part of the physical source ledger.

Proposition 4.1 (Conditional rotor consequences)**.** *Assume Eq. (4.1), a positive branch $n > 0$, and the infrared constitutive functions*

$$\varepsilon(n) = m_R n + \frac{u_R}{3} n^3, \qquad \mathcal{K}(y) = \frac{\kappa_R}{3} y^3, \qquad m_R, u_R, \kappa_R > 0. \tag{4.5}$$

Then:

(i) phase variation gives the exact finite repair-current balance Eq. (4.4);

(ii) the homogeneous dilute phase has

$$\rho_R = m_R n + \frac{u_R}{3} n^3, \qquad p_R = \frac{2u_R}{3} n^3, \tag{4.6}$$

so source-free expansion gives $n \propto a^{-3}$, $\rho_R \propto a^{-3}$, and $w_R \simeq 0$ when $u_R n^2 \ll m_R$;

(iii) if $\mathcal{Q}_b = \beta_b \rho_b$ and $\mathbf{a}_R = -\beta_b \nabla\theta$, the static spherical exterior branch gives

$$a_R = \sqrt{a_b a_0}, \qquad v^4 = G M_b a_0, \qquad a_0 = \frac{\beta_b^3}{4\pi G \kappa_R}; \tag{4.7}$$

(iv) translation of the coherent source gives Eq. (4.9) and the momentum ledger Eq. (6.2).

These statements are consequences of the proposed action, not derivations of that action from the finite OPH theorems.

Proof. Item (i) follows by varying θ and summing the local equation over a finite region. For item (ii), the first-order rotor Hamiltonian density is $\varepsilon(n)$, and its barotropic pressure is $p_R = n\varepsilon'(n) - \varepsilon(n)$. Source-free homogeneous continuity then gives $na^3 = \text{constant}$. For item (iii), spherical integration of $\nabla \cdot (\kappa_R |\nabla\theta| \nabla\theta) = \beta_b \rho_b$ gives $4\pi r^2 \kappa_R |\theta'| \theta' = \beta_b M_b$; the stated acceleration and definition of a_0 give the result. Item (iv) follows by translating the source term in Eq. (4.1) and applying translation invariance to matter plus repair field. $\square$

Define the integrated device repair charge

$$Q_R^{\text{dev}} = q_\star \chi_\nu^{\text{can}} \int_\Omega S_{\text{coh}}^{\text{can}}(x)\, \mathrm{d}^3 x. \tag{4.8}$$

For a prescribed external repair phase, translation of the source gives

$$\boxed{F_i^{\text{coh}} = -q_\star \chi_\nu^{\text{can}} \int_\Omega S_{\text{coh}}^{\text{can}}(x)\, \partial_i \theta_{\text{ext}}(x)\, \mathrm{d}^3 x.} \tag{4.9}$$

If the external gradient varies slowly across the device,

$$\boxed{F_i^{\text{dev}} \simeq -Q_R^{\text{dev}} \partial_i \theta_{\text{ext}}.} \tag{4.10}$$

The sign of Q_R^{dev} may select attractive or repulsive response. Positive repair energy is compatible with signed charge when $\varepsilon(n) = \varepsilon(|n|)$.

Remark 4.2 (Switchable-sign boundary)*.* The gated source in Eq. (1.1) is nonnegative under its displayed definition. With fixed $q_\star$ and positive χ_ν^{can}, exchanging top and bottom source profiles does not reverse the monopole Q_R^{dev}. A switchable force sign therefore requires either a signed orientation factor derived as part of the coherent source map or a controlled reversal of $\nabla\theta_{\text{ext}}$. The finite channel theorem supplies neither operation.

5 Compact-Source Constraint

Proposition 5.1 (No monopole from a neutral compact source)**.** *Suppose the coherent repair source is compactly supported and satisfies*

$$\int_\Omega \mathcal{Q}_{\rm coh}\,\mathrm{d}^3x = 0. \tag{5.1}$$

Then it has no repair monopole. Its exterior field begins at dipole order or higher. In a uniform external gradient, the leading forces on its positive and negative charges cancel.

Proof. Integrating Eq. (4.3) in a static state makes the exterior flux equal the integrated source. Equation (5.1) sets the monopole flux to zero. The multipole expansion therefore begins beyond the monopole. A uniform external gradient couples only to total charge at leading order, which vanishes. □

A bottom-to-top contrast

$$\Delta S = S_{\rm bottom} - S_{\rm top} \tag{5.2}$$

can shape a dipole, select the sign of injected repair charge, couple to a nonuniform external field, or drive an alternating repair current. It does not by itself define a nonzero $Q_R^{\rm dev}$. No formula proportional only to $g^2 A\chi_\nu \Delta S/(4\pi G)$ is a force theorem for a closed compact device.

6 Momentum and Energy Closure

For the cubic condensed branch, the repair-field stress is

$$T^R_{ij} = \kappa_R |\nabla\theta|\,\partial_i\theta\,\partial_j\theta - \delta_{ij}\frac{\kappa_R}{3}|\nabla\theta|^3 + \cdots\,. \tag{6.1}$$

The exact force ledger is

$$\boxed{F_i^{\rm matter} + \frac{\mathrm{d}P_i^R}{\mathrm{d}t} + \oint_{\partial\Omega} T^R_{ij} n^j\,\mathrm{d}A = 0.} \tag{6.2}$$

A local repulsive response requires either nonzero device repair charge with a field tail or explicit momentum transfer through Eq. (6.2). A static, closed, repair-neutral device cannot lift itself.

Any switchable force also requires a toggle-energy ledger. Let $E(q,s)$ be the energy at position q and internal state s, and define

$$\mathrm{Toggle}(q) = E(q,\mathrm{on}) - E(q,\mathrm{off}). \tag{6.3}$$

For an on/off spatial cycle between q_1 and q_2, conservation gives

$$W_{\rm cycle} = \mathrm{Toggle}(q_1) - \mathrm{Toggle}(q_2). \tag{6.4}$$

A claimed position-dependent switchable force with no corresponding toggle or field-energy difference violates the closed energy ledger.

7 Laboratory Evidence Conditions

A force experiment has physical standing only if it supplies:

1. an instrument-independent measurement of $S_{\rm coh}^{\rm can}$ or its declared engineering coordinate;
2. the dimensional calibration $q_\star$;
3. a measured ambient repair gradient or a controlled emitted repair field;
4. a nonzero integrated repair charge, or a measured outgoing repair-current and momentum flux;
5. a complete toggle, maintenance-power, and field-energy ledger;
6. detuned, dummy, reversed-sign, thermal, acoustic, electromagnetic, mechanical, and center-of-mass controls;
7. a force result that follows the repair-field geometry and sign law in Eq. (4.9).
8. for any claimed switchable reversal, a derived signed source factor or an independently measured reversal of the external repair gradient.

Without these receipts, a null result does not measure $\chi_\nu^{\rm can}$, and a nonzero balance signal does not establish repair charge. A test of $\chi_\nu^{\rm can}$ requires the action, source calibration, and operational source observable listed above.

8 Conclusion

The canonical coherent-matter susceptibility is an order-one channel coefficient fixed by protected-reserve survival on the stated OPH branch. The repair-charge condensate completion gives that coefficient a lawful dynamical role: it weights the coherent source in the repair-current equation. The same proposed action yields exact finite continuity, a dilute dust limit, the spherical deep-galaxy scaling, repair-field stress, and a device-force law. A device force depends on the dimensional calibration, integrated repair charge, external field, and field-momentum ledger. Repulsive response is conditionally allowed. Reactionless lift from a compact neutral contrast is excluded.

References

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